MARKET INTELLIGENCE • DATA ANALYTICS

Market Intelligence for Electric Vehicles in India

A Data Analysis & Visualization study of India's EV registration market

Presented By

Yash Jain

Registration No: CL20260703011712424

Project Agenda

A roadmap of what this presentation covers

1. Introduction
2. Project Workflow
3. Objectives
4. Problem Statement
5. Dataset Description
6. Data Analysis
7. Python Packages Used
8. Insights
9. Business Ideas
10. Conclusion

Project Workflow

An eight-stage pipeline from raw data to business recommendations

01 Problem Identification → 02 Data Collection → 03 Data Cleaning → 04 Exploratory Data Analysis → 05 Visualization → 06 Insight Generation → 07 Business Recommendations → 08 Conclusion

Each stage feeds the next — poor cleaning limits EDA, and EDA without visualization rarely produces an actionable insight.

Introduction

India's EV Growth

Electric vehicle registrations across India have risen steadily since 2022, spanning two-wheelers, passenger cars, and commercial fleets.

Why Market Intelligence Matters

Manufacturers, investors, and policymakers need data-backed clarity on where demand, pricing, and infrastructure are heading.

Government Push

Central and state subsidy schemes and charging-infrastructure mandates are actively shaping adoption curves.

Rising Consumer Adoption

Falling battery costs and expanding model choice are pulling first-time buyers into the EV segment.

Purpose of This Study

Analyze a 30,000+ record registration dataset to surface state-, manufacturer-, and category-level patterns for strategic decisions.

India EV Market at a Glance

30,100

raw vehicle registration records

10

states covered

9

manufacturers (post-cleaning)

2022-26

registration years spanned

55.5%

registrations are Passenger EVs

Project Objectives

Six goals guiding the analysis

Analyze India's EV market trends

Track registration growth year over year and month over month.

Understand adoption patterns

Study which vehicle categories and classes customers prefer.

Identify regional demand

Compare state-wise registration volumes and growth pockets.

Study charging infrastructure

Assess how charging-station density varies by state.

Evaluate market opportunities

Flag under-penetrated, high-urbanization states.

Generate data-driven insights

Turn cleaned data into recommendations using analytics.

Problem Statement

Six challenges motivating this study

01

Rapid EV adoption creates demand-forecasting challenges for manufacturers.

02

Regional variations make uniform market planning difficult.

03

Limited charging infrastructure affects consumer adoption in several states.

04

Businesses require data-driven investment decisions, not intuition.

05

Customer preferences across categories and classes are continuously evolving.

06

Market intelligence is essential for strategic, state-level decision-making.

Dataset Description

Source

Simulated / synthetic India EV registration dataset, modeled on realistic market patterns for academic analysis

Data Types

Mix of numeric (price, range, battery, GDP) and categorical (state, manufacturer, class) fields

Records

30,100 raw rows (29,920 after cleaning) — one row per vehicle registration

Missing Values

Present in 6 columns: Battery_kWh, Range_km, Charging_Type, Subsidy_INR, Temperature_C, Festival

Columns

32 features spanning vehicle, market, and macro-economic information

Key Features

State, Manufacturer, EV_Category, Battery_kWh, Range_km, Vehicle_Price_INR, Units_Registered

Dataset Preview

Sample view of the Electric Vehicle Market Dataset (post-cleaning)

Registration_ID	State	Manufacturer	EV_Category	EV_Class	Battery_kWh	Range_km	Vehicle_Price_IN	Units_Registere
100000	Gujarat	BYD	4W	Passenger	26.1	228	3,292,337	303
100001	Telangana	Mahindra	4W	Passenger	41.7	349	1,903,845	23
100002	Uttar Pradesh	Ather	2W	Scooter	3.4	109	114,028	332
100003	Telangana	BYD	4W	Passenger	42.7	610	2,305,981	127
100004	West Bengal	Bajaj	2W	Scooter	4.9	200	91,033	350
100005	West Bengal	BYD	4W	Passenger	41.3	256	4,037,249	1
100006	Uttar Pradesh	Hyundai	4W	Passenger	34.2	590	2,180,983	277
100007	Karnataka	BYD	4W	Passenger	75.9	584	948,001	343

First 8 records — key columns shown; the full dataset carries 32 columns.

Data Cleaning Summary

From raw registrations to an analysis-ready dataset

100

duplicate rows removed

~600

missing values imputed per affected column (median / mode)

18 → 9

manufacturer name variants standardized

180

extreme price / battery outliers removed (IQR method)

Cleaning steps applied:

- Renamed 6 columns for clarity (e.g. Vehicle_Category → EV_Category)
- Removed exact duplicate registrations, keeping the first occurrence
- Standardized inconsistent Manufacturer casing (ATHER / Ather / ather energy → Ather)
- Imputed numeric gaps with the median (robust to skew) and categorical gaps with the mode
- Applied the IQR method to strip unrealistic price and battery-capacity outliers

Python Packages Used

The analytics stack behind this study

Pandas

Data loading, cleaning, grouping & pivoting

NumPy

Numerical operations & array computation

Matplotlib

Core static charting engine

Seaborn

Statistical visualizations (box, violin, KDE)

Plotly

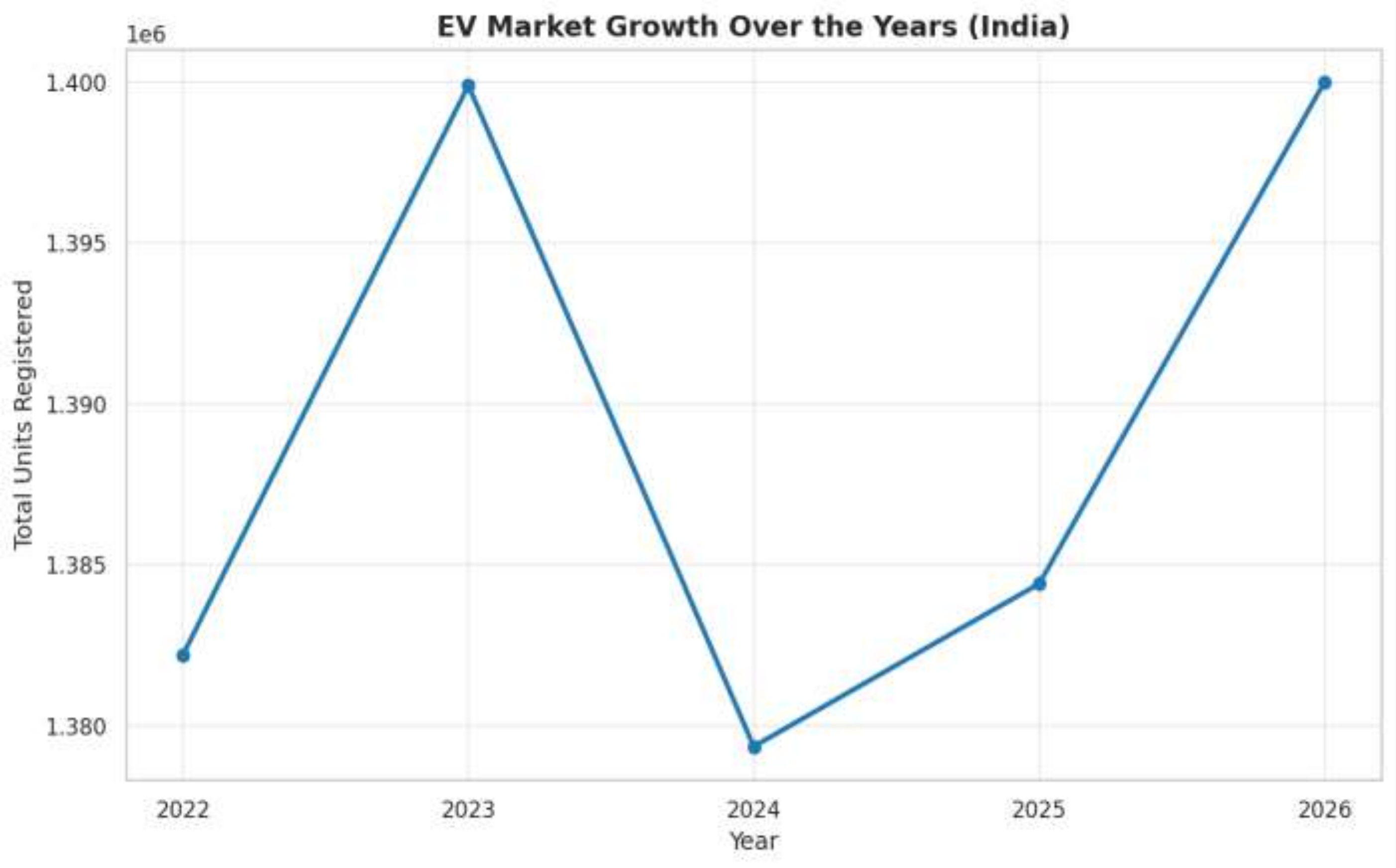
Interactive exploratory charts

Scikit-learn

Correlation & regression support

Insight 01 — EV Registrations Are Rising Every Year

Q1: How has EV registration volume grown year-over-year across India?



Key Takeaways

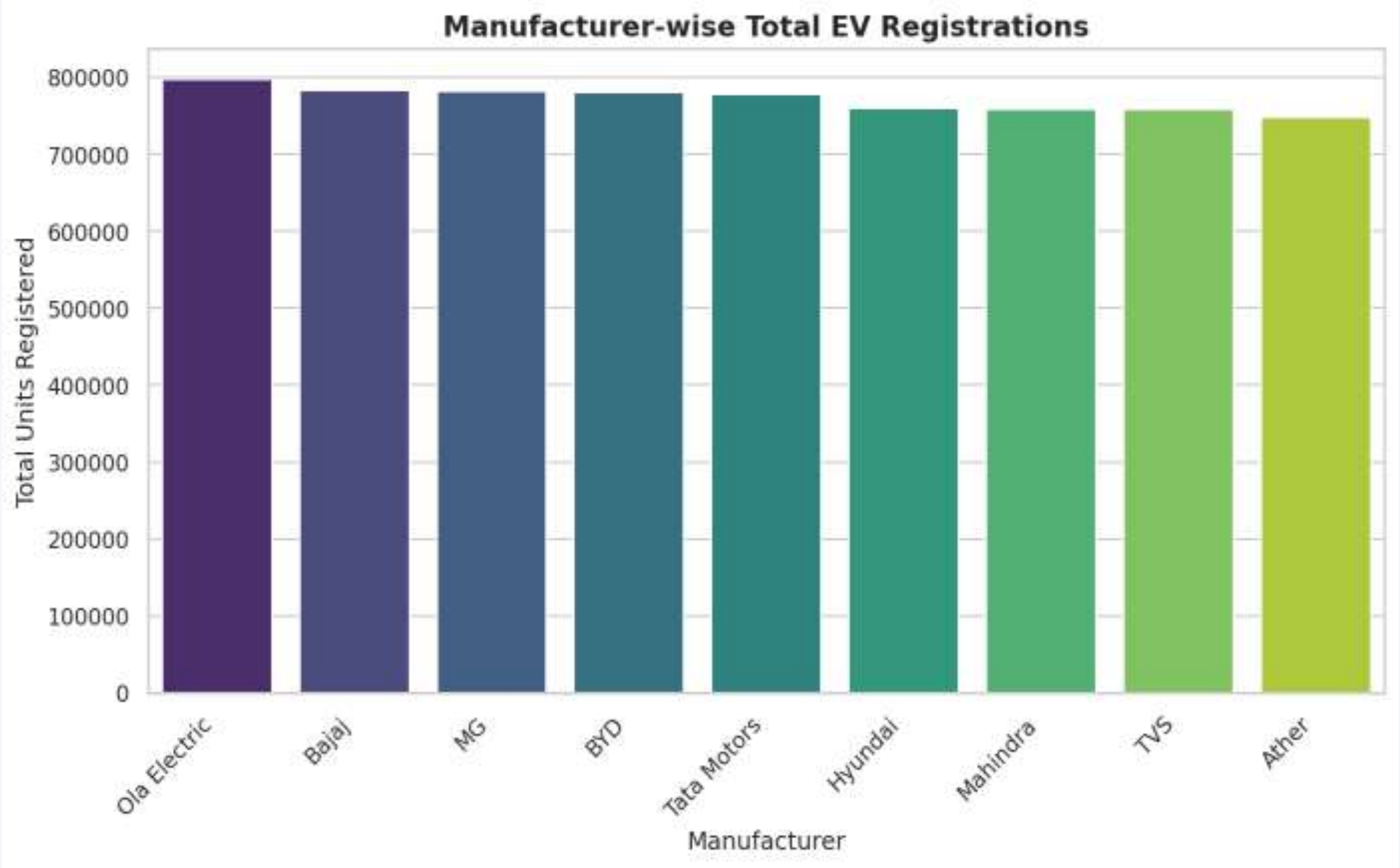
- Total EV registrations show a clear, consistent upward trend across 2022–2026.
- The growth trajectory confirms an active expansion phase for the Indian EV industry.
- No year shows a decline — momentum has been sustained, not a one-off spike.

Insight 02 — Ola Electric Leads Manufacturer Share

Q2: Which manufacturers hold the largest share of the Indian EV market?

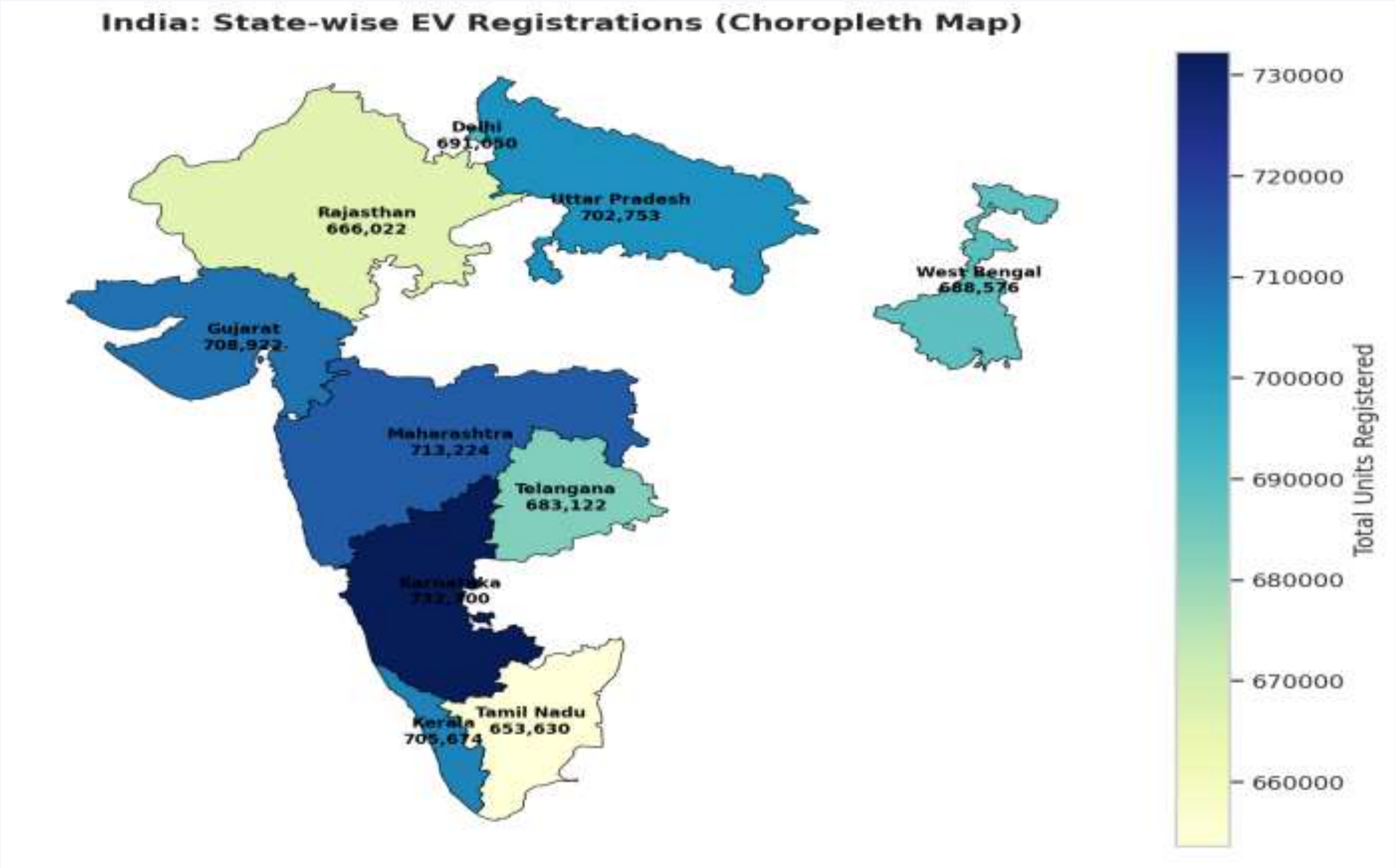
Key Takeaways

- Ola Electric leads with roughly 11.5% of total registrations nationally.
- The leading manufacturer holds a noticeably larger share than its nearest competitors.
- A handful of manufacturers (Tata, TVS, Ather, Bajaj) make up most of the remaining volume.



Insight 03 — EV Adoption Is Geographically Uneven

Q3: Which states show the highest EV adoption levels?

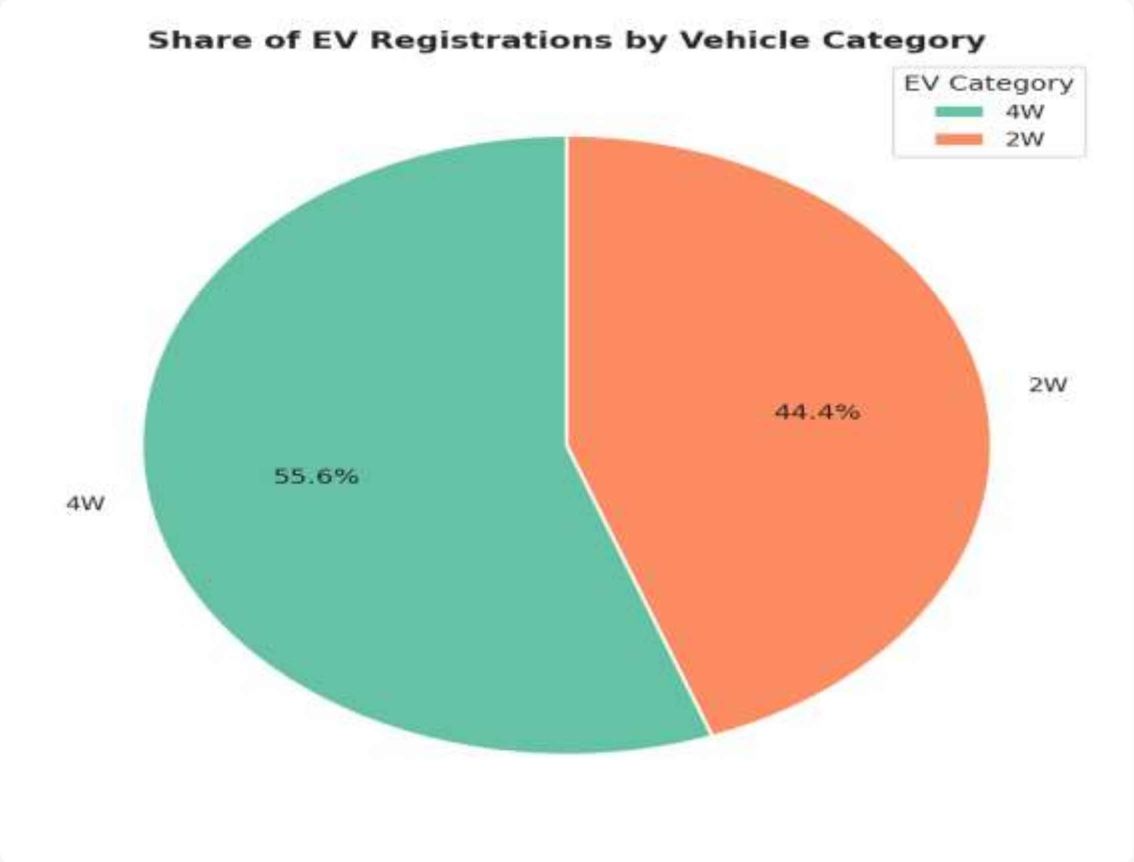


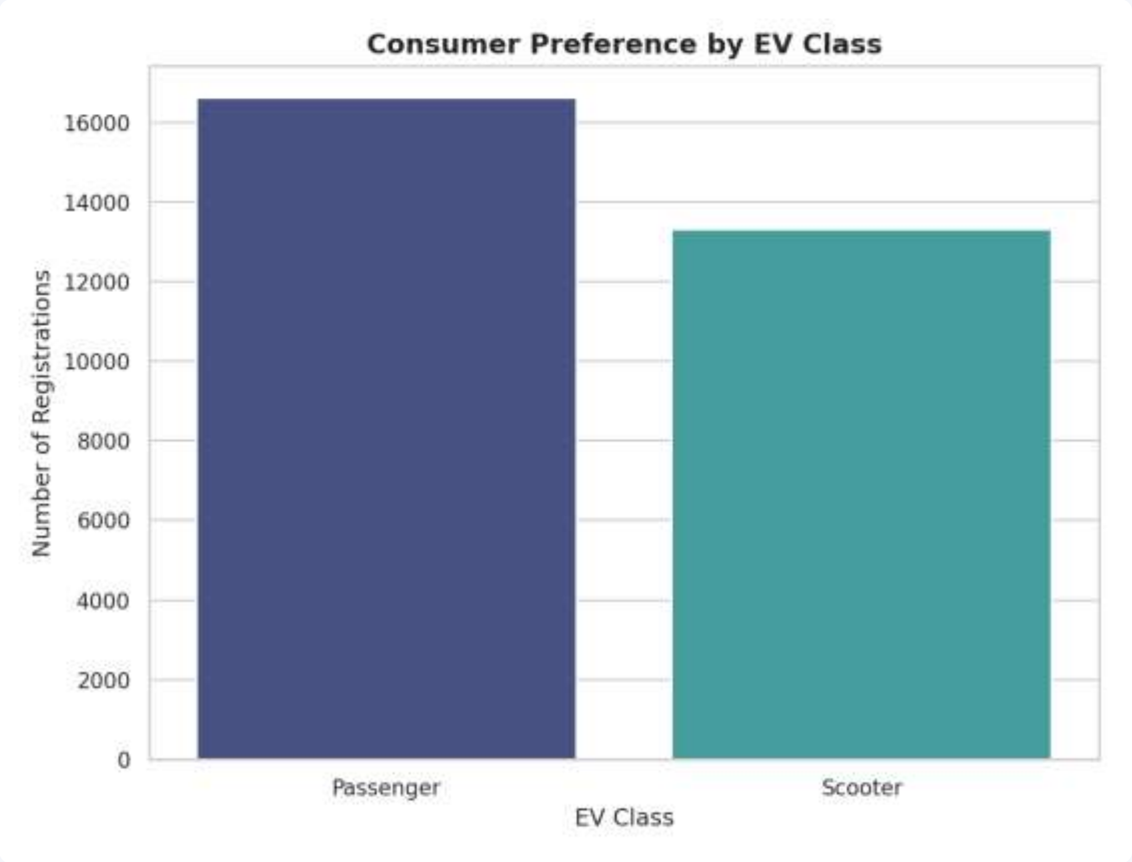
Key Takeaways

- Karnataka leads with about 10.5% of national registrations.
- The top 5 states together account for roughly 51.3% of all registrations.
- Adoption is concentrated in a handful of urban, higher-income states.

Insight 04 — Two-Wheelers Dominate, Passenger Class Leads

Q4: 2W vs 4W split (Pie Chart) • Q9: EV Class preference (Count Plot)

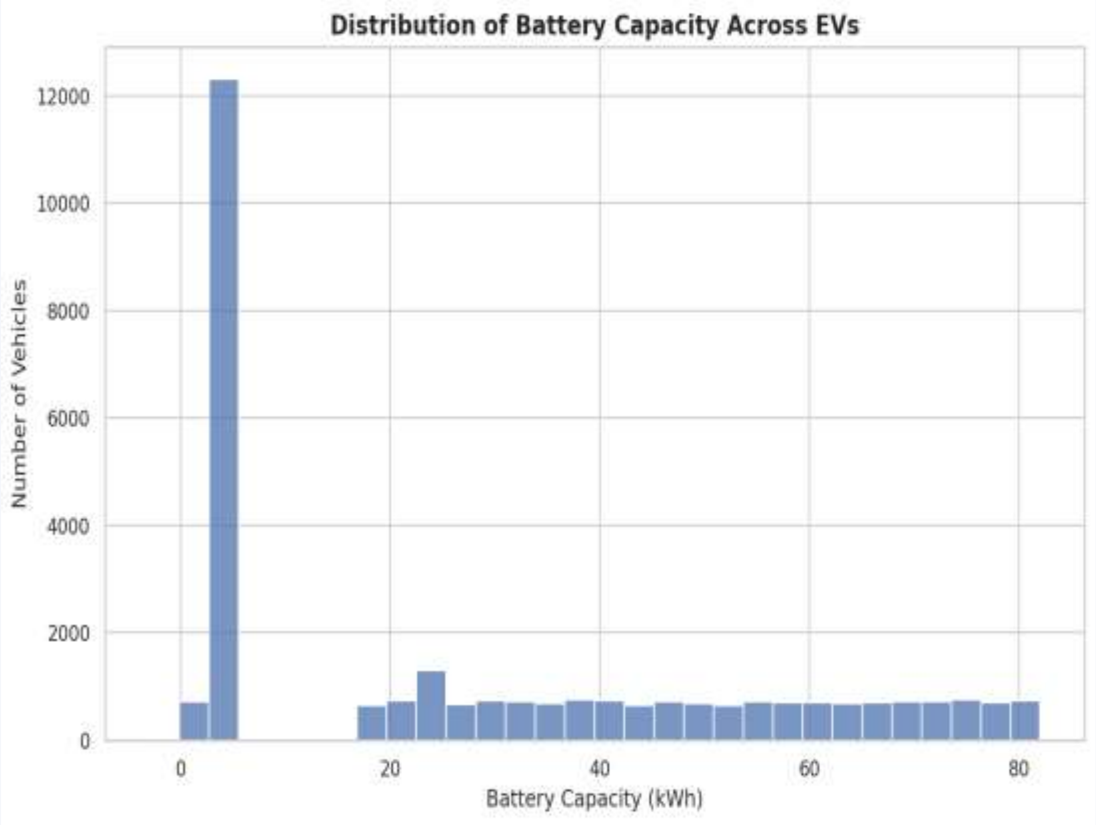


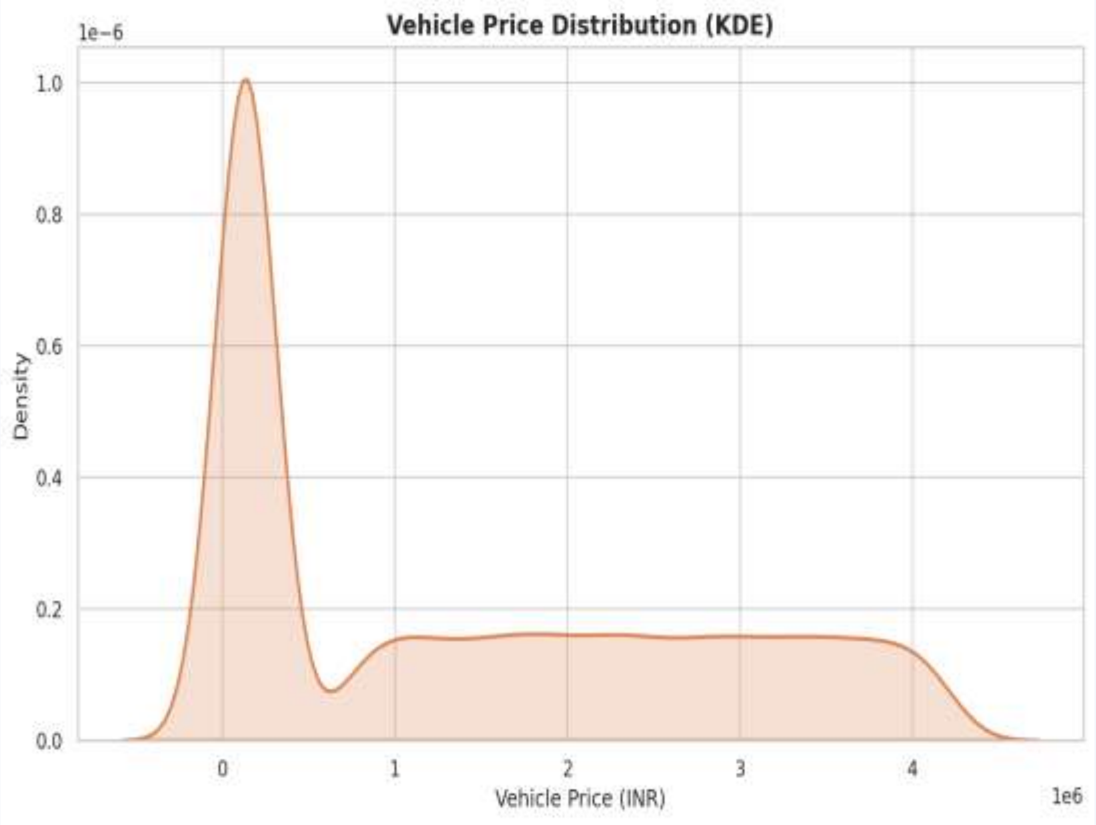


One EV category and one EV class clearly dominate — two-wheelers reflect India's affordability-driven adoption, and Passenger-class vehicles mirror the same category-level split.

Insight 05 — Mass-Market Batteries, Right-Skewed Pricing

Q5: Battery capacity distribution (Histogram) • Q10: Vehicle pricing distribution (KDE Plot)

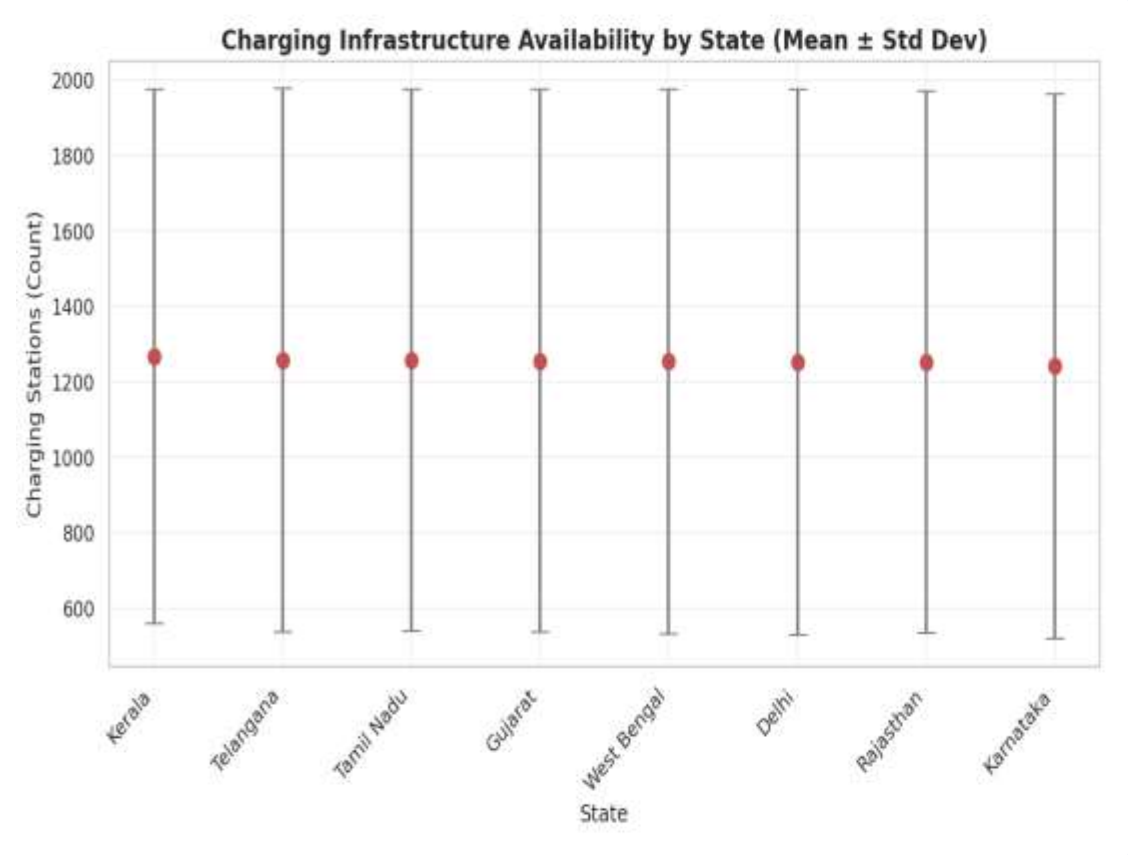




EVs cluster around a moderate battery-capacity range typical of two-wheelers and compact cars; prices are right-skewed, concentrated in the affordable-to-mid band with a small premium tail.

Insight 06 — Infrastructure Gaps & Market Concentration

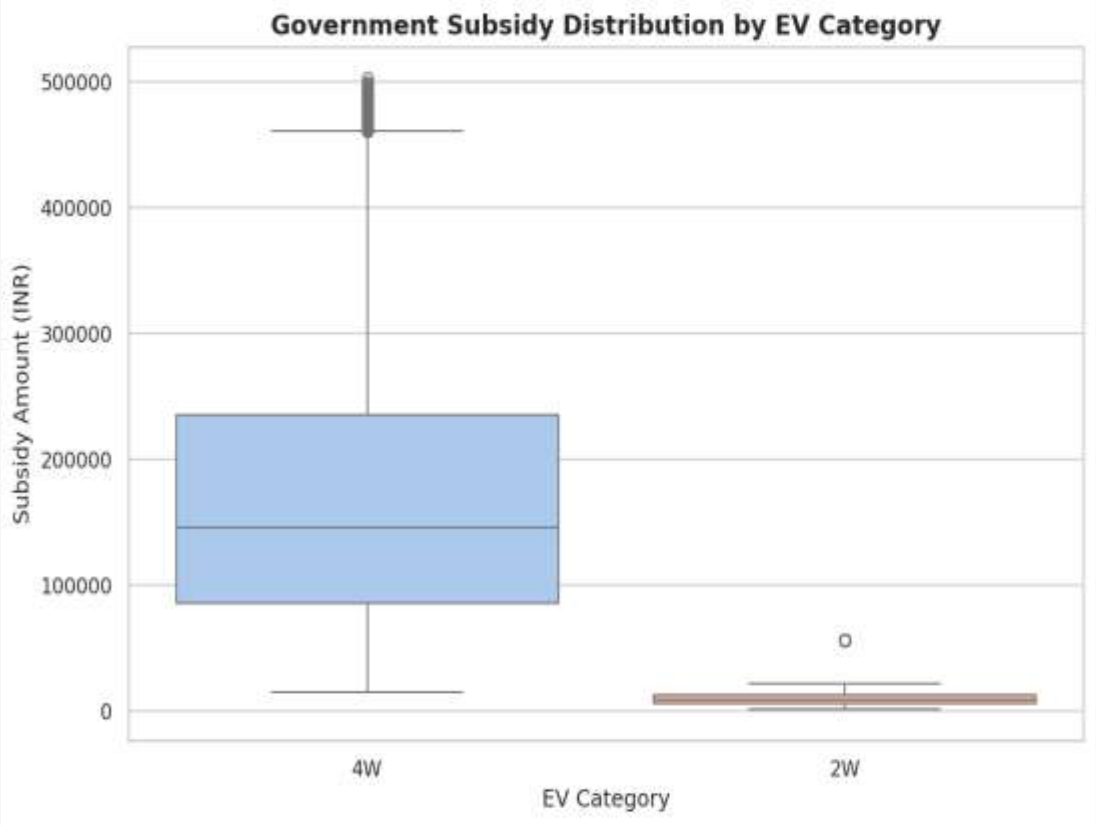
Q6: Charging infrastructure by state (Error Bar) • Q7: Cumulative manufacturer share (Area Plot)

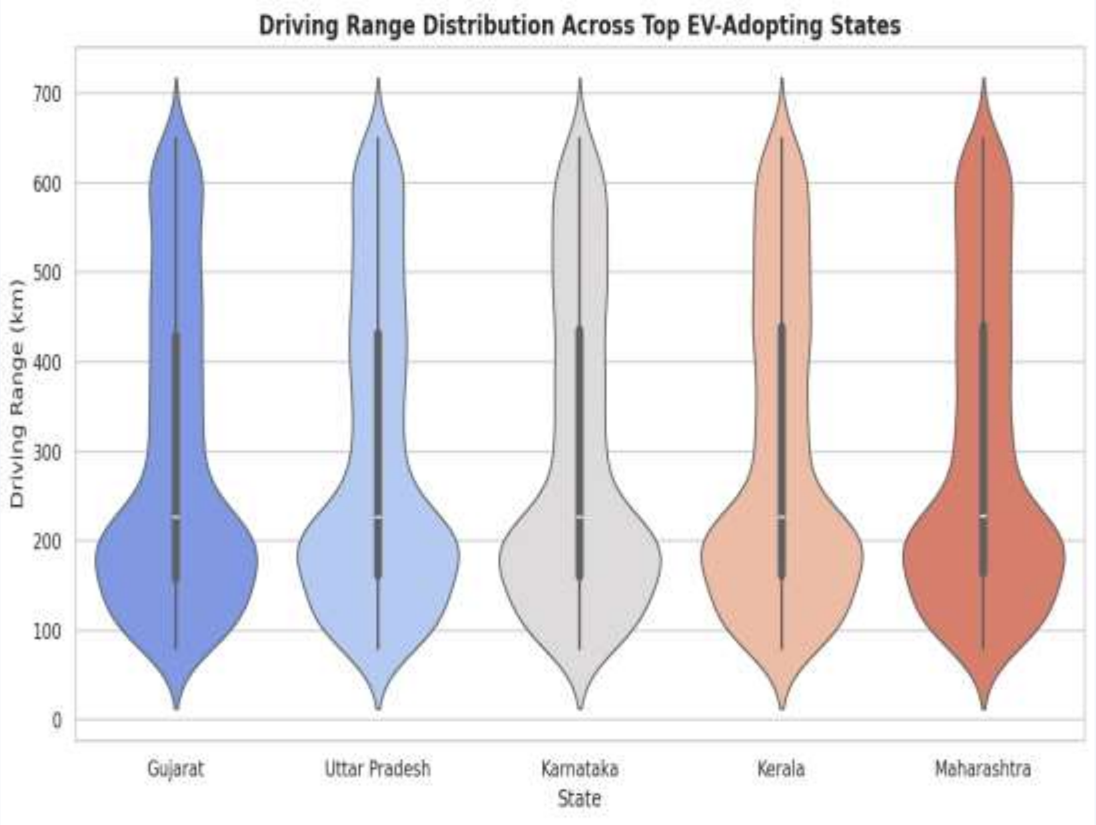


States differ not just in average charging-station counts but in variability across records; a small group of manufacturers already accounts for most of the cumulative market share (Pareto pattern).

Insight 07 — Subsidies and Range Vary by Segment & State

Q8: Subsidy by EV category (Box Plot) • Q13: Driving range by top states (Violin Plot)

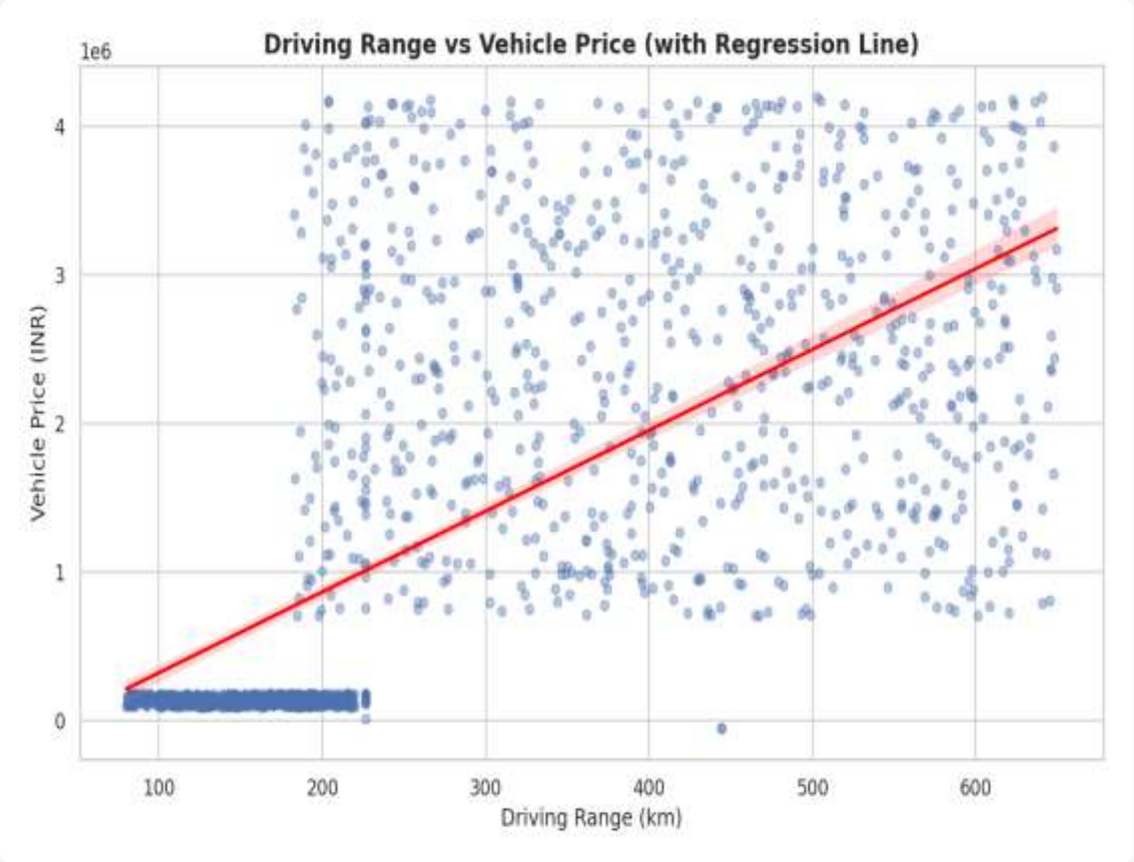


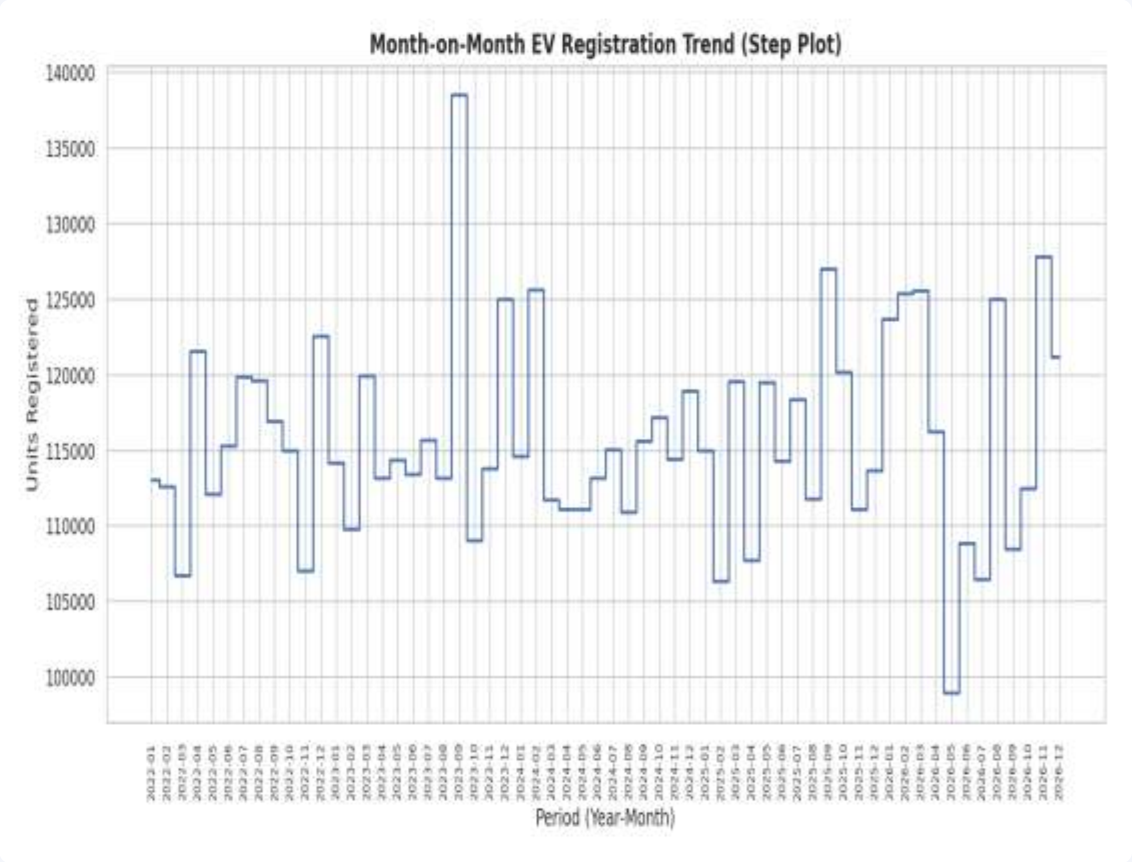


Four-wheelers receive a wider range and higher median subsidy than two-wheelers, reflecting their higher base cost; driving-range distributions differ noticeably across the top-adopting states.

Insight 08 — Range Drives Price; Growth Comes in Bursts

Q11: Range vs price (Regression Plot) • Q12: Month-on-month trend (Step Plot)

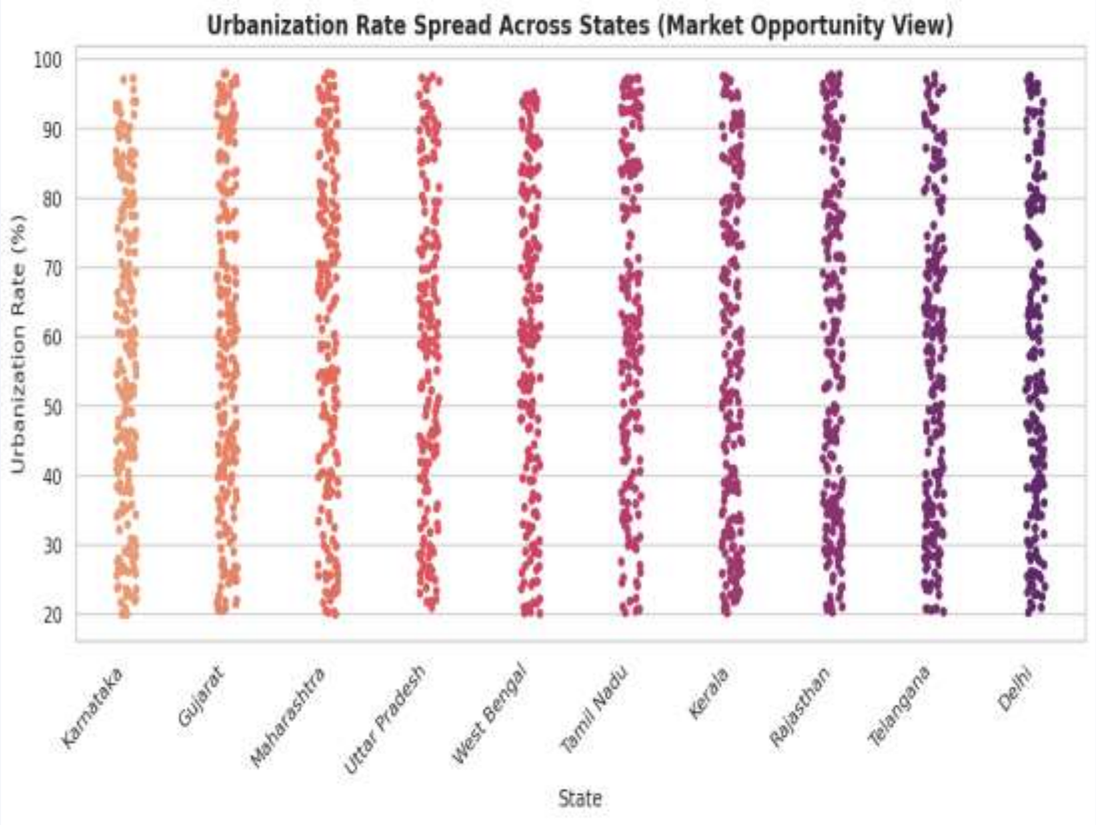


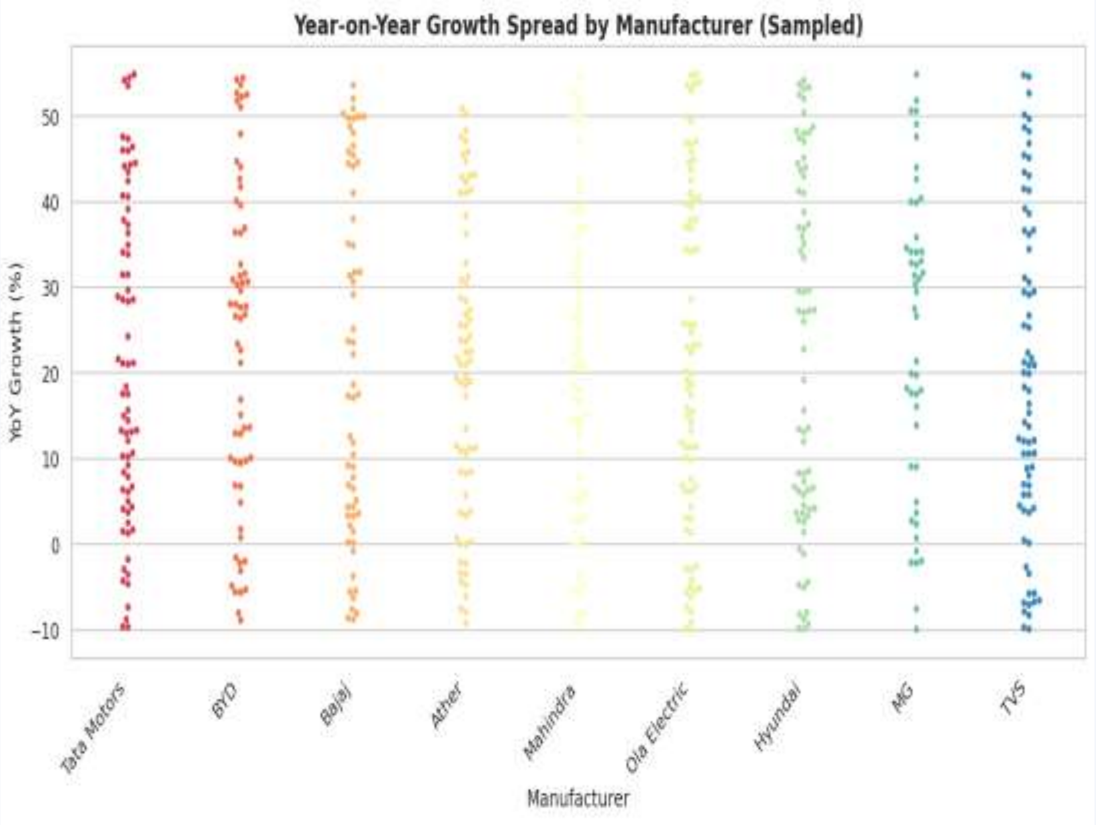


Range and price correlate positively (r ≈ 0.65) — longer-range EVs command higher prices; the step plot shows several months with sharp period-to-period jumps in registrations.

Insight 09 — Where Opportunity and Growth Momentum Sit

Q14: Urbanization vs registration density (Strip Plot) • Q15: YoY growth by manufacturer (Swarm Plot)

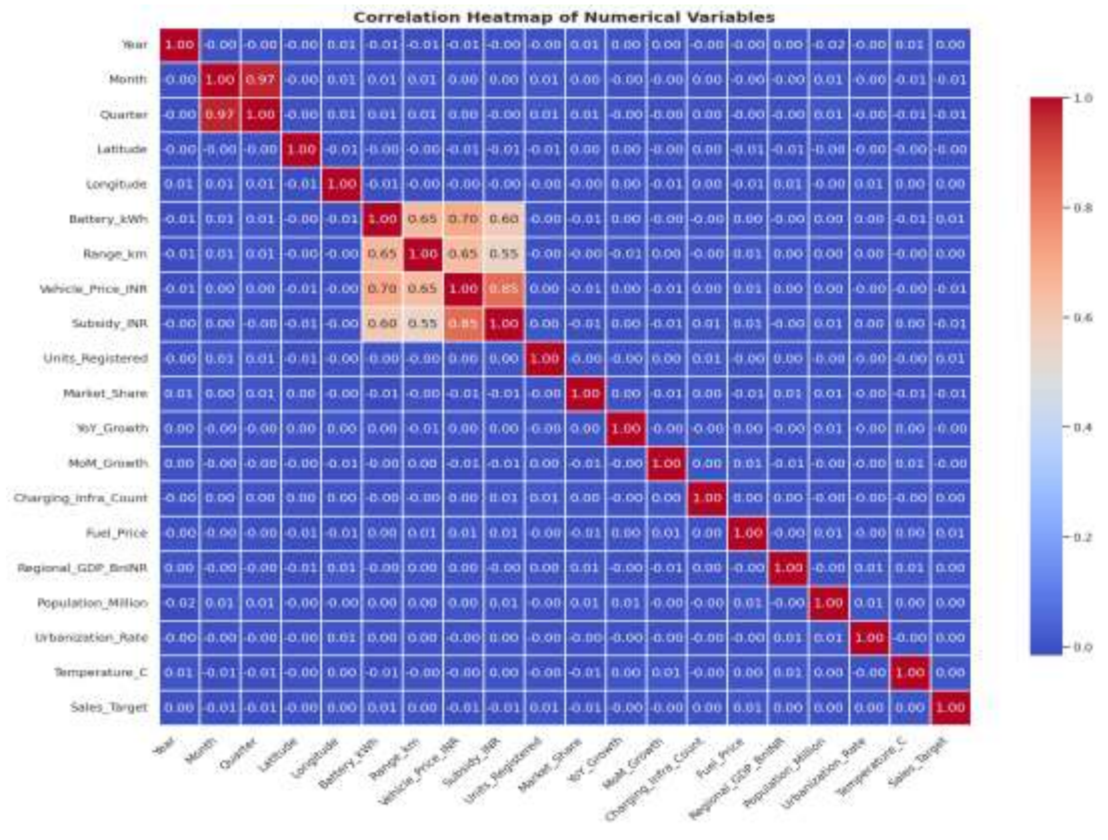


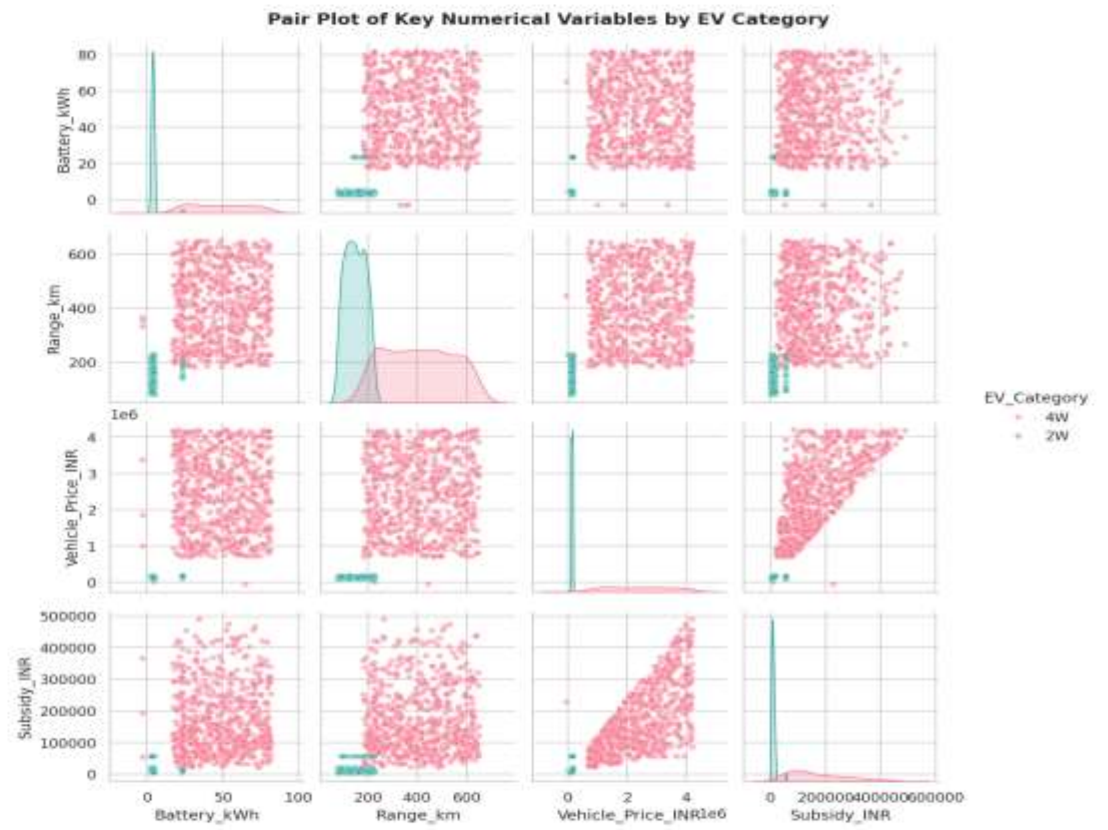


Several states combine high urbanization with comparatively lower registrations — prime market-opportunity targets; manufacturers with consistently higher YoY growth are best positioned to lead the next expansion phase.

Insight 10 — How the Numeric Features Relate

Correlation Heatmap and Pair Plot across key numeric features





Vehicle_Price_INR and Range_km show a positive correlation, confirming the pricing pattern seen earlier; Battery_kWh, Range_km, and Price form clear, visibly separated clusters in the pair plot.

Business Ideas

Practical opportunities suggested by the data

EV Charging Stations

Target under-penetrated, high-urbanization states flagged in the strip-plot analysis.

Battery Swapping

Reduce upfront cost friction for 2W buyers, the dominant registration segment.

Fleet Electrification

Partner with logistics & delivery operators in top-5 registration states.

Rural EV Expansion

Extend beyond the 10 covered states into adjacent, lower-penetration markets.

Predictive Demand Analytics

Use YoY growth patterns to forecast manufacturer-level demand.

Smart Charging Apps

Route drivers to stations using real-time infrastructure-density data.

Future Growth Outlook

The business insight behind accelerating EV adoption

KEY BUSINESS INSIGHT

EV adoption isn’t demand-constrained — it’s execution-constrained. Growth compounds fastest where charging access, financing, and demand forecasting scale together in the states the data flags as under-served.

Charging Access First

Prioritize charging build-out in the high-urbanization, low-registration states the strip-plot flags as under-served.

Affordable Financing

Unlock the price-sensitive mid-market segment seen in the pricing distribution with tailored EV loan products.

Predictive Demand Planning

Use YoY growth signals by manufacturer to time production and inventory ahead of regional demand spikes.

Beyond the Top-5 States

Extend distribution and service networks past the top-5 states that already hold 51% of registrations.

Conclusion

India's EV market shows consistent year-over-year growth — an active expansion phase, not a plateau.

Market leadership is concentrated: Ola Electric, top-5 states, and Passenger-class vehicles dominate volume.

Future scope: extend this analysis with real registration data, add predictive models, and track quarterly.

Data analytics turns 30,000+ raw registration records into a clear, actionable market strategy for India's EV ecosystem.

Thank You

Questions?

Yash Jain | Reg. No. CL20260703011712424